

SCIALOG: Temporal & Sequence-Based Behavioral Analysis

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(Core analyses complete; refinement ongoing)

Study Description: Using the codebook synthesized from existing literature reviews, we constructed temporal and sequence-based behavioral features to examine whether when and how behaviors unfold predict team roles.

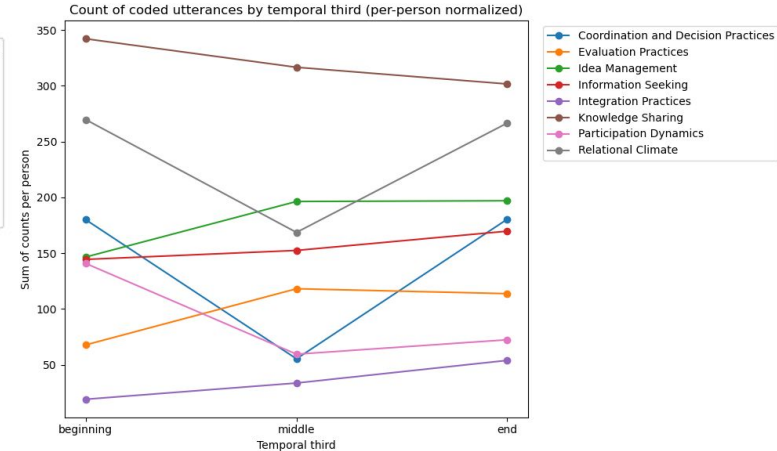
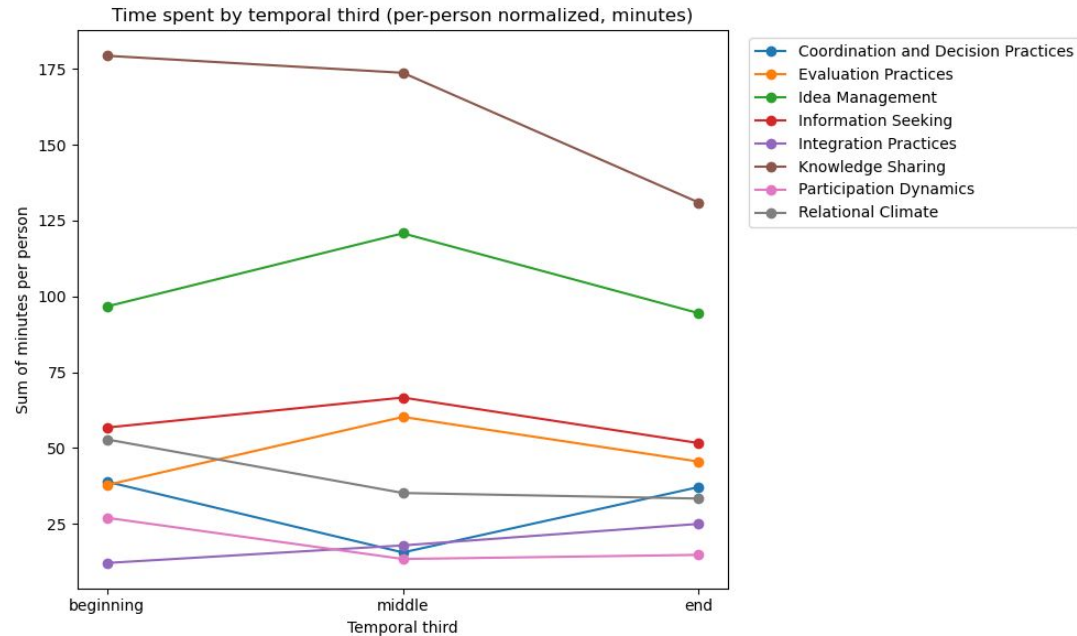
Study Questions:

1. Do early meeting behaviors predict later roles and outcomes?
2. Do facilitators, team members, and funded participants exhibit distinct temporal patterns?

Data Sample:

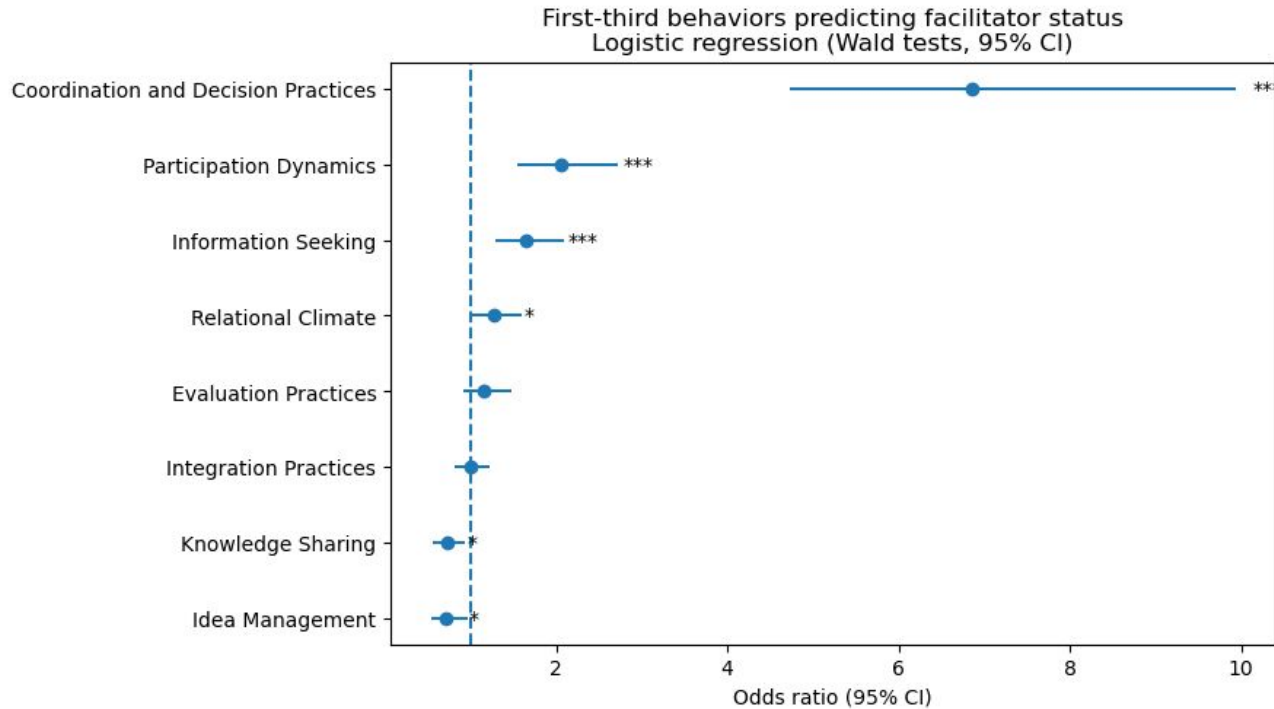
- 128 SCIALOG meeting recordings (1-1.5 hrs with 6-18 participants)
- Gemini + human-validated behavioral annotations (aligned with the meetings split into thirds)
- Three outcomes: (1) facilitator role, (2) team membership, (3) funding status

Temporal Behavioral Patterns Across All SCIALOG Meetings



Knowledge Sharing is front-loaded, while Idea Management peaks mid-meeting, suggesting early grounding and mid-phase structuring. Relational and participation behaviors dip mid-meeting before partially recovering towards the end

Which First ⅓ of a Meeting Behaviors Distinguish Facilitators From Non-Facilitators?

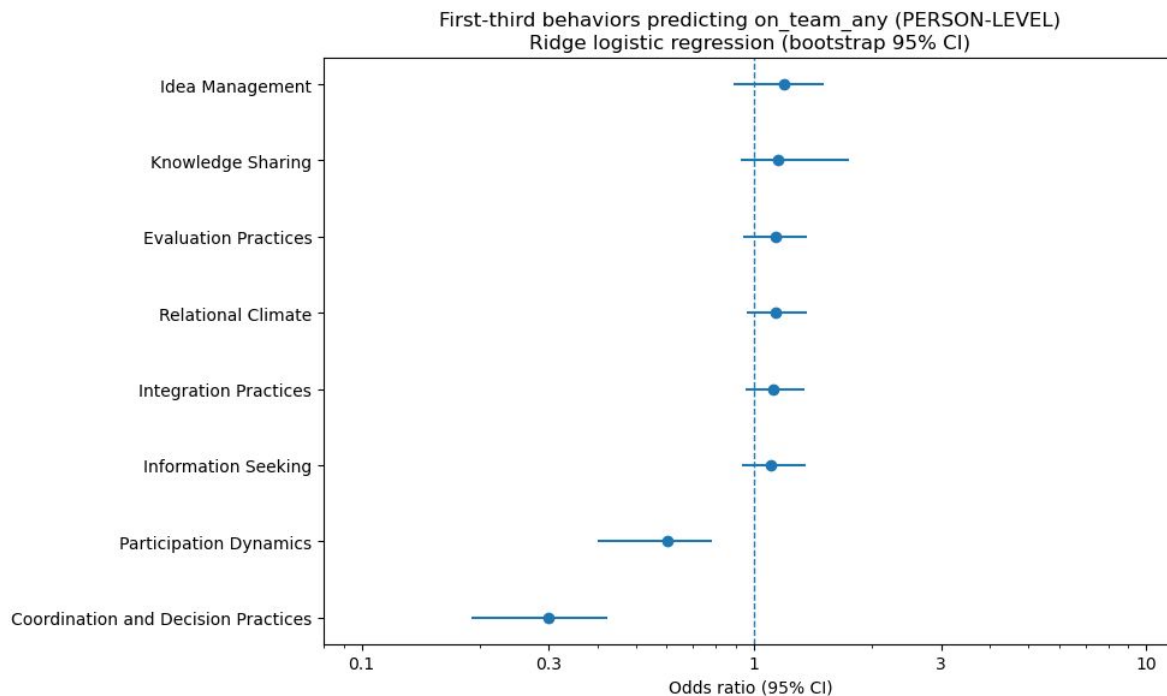


Facilitators exhibit higher early Coordination, Participation, Information Seeking, and Relational Climate

Higher Idea Management and Knowledge Sharing are less characteristic of facilitators

Features represent summed speaking time (minutes) per code annotation

Meeting Behaviors in the First 1/3 of a Meeting Differentiate Team Formation



summation of time per annotation code @ individual level

Higher early Participation Dynamics and Coordination & Decision Practices are associated with lower odds of team formation

Other early behaviors show weaker or inconsistent associations

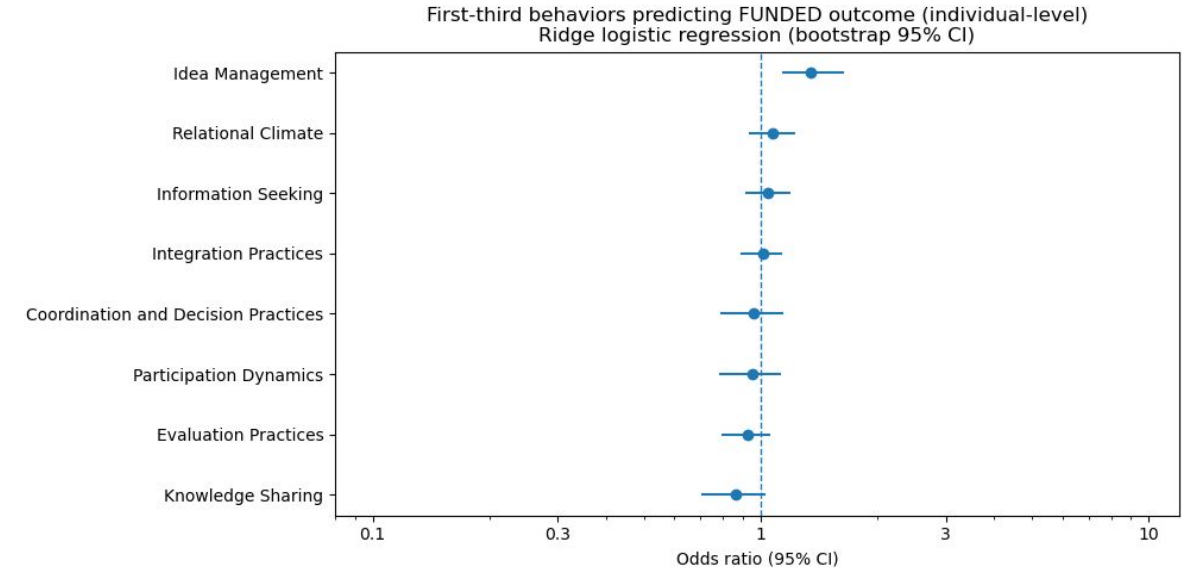
In General:

- team formation (more people on a team)
- team funding success (more balanced)

successful/unsuccessful proposals that have significant differences (5% sig differences)

- what distinguishes the most successful teams from unsuccessful ones (and how big is that effect)
- Keep this graph. if i go back to check (count/time) it would be nice to have (see EV graph photo)

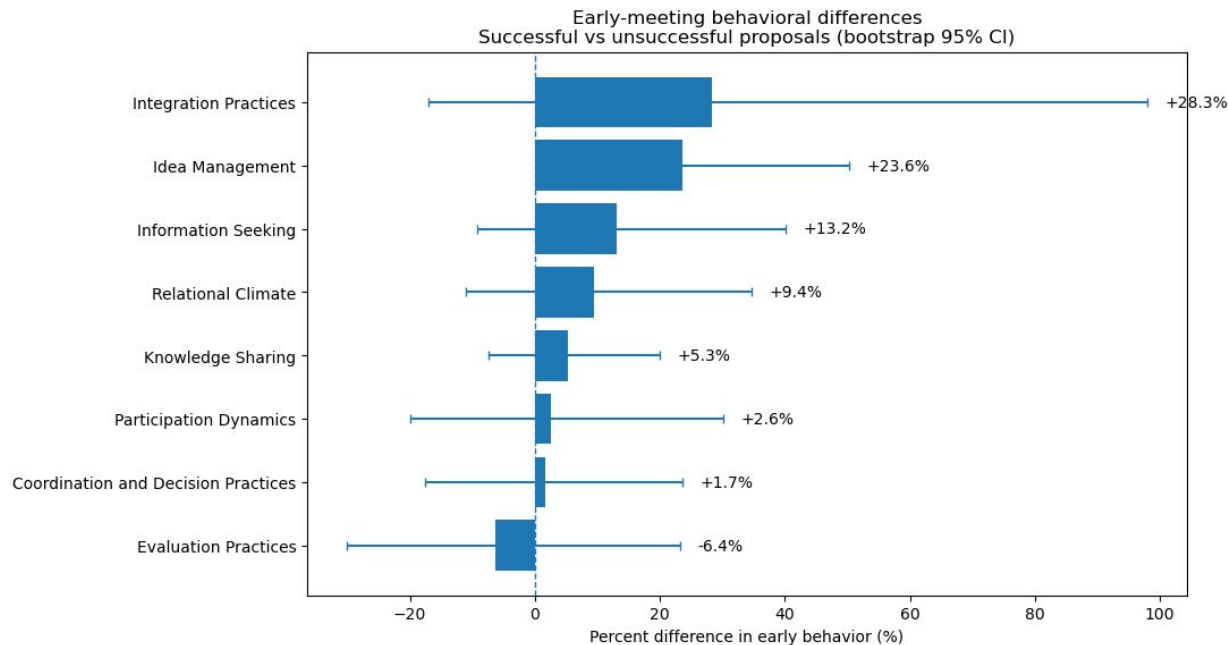
First 1/3 of a Meeting Behaviors and Funding Outcomes



At the individual level, only Idea Management shows a reliable association with funded outcomes

Other early behaviors show no reliable early association with funding outcomes

First 1/3 of a Meeting Behavioral Differences Between Successful and Unsuccessful Proposals



Percent difference = $100 \times (\text{mean_successful} - \text{mean_unsuccessful}) / |\text{mean_unsuccessful}|$

Unsuccessful proposals are characterized by higher levels of early Evaluation, Integration, and Coordination & Decision Practices

The largest differences appear in Evaluation, Integration Practices ($\approx 20-30\%$)

What Signals from the **First 1/3 of Meetings** Reveal

1. Early meeting behaviors already differentiate roles, team formation, and funding outcomes
2. The strongest early signals reflect idea-focused behaviors (Knowledge Sharing, Information Seeking, etc.) not the raw participation volume
3. Temporal positioning of behaviors suggests structured idea flow early in meetings, motivating a linkography-inspired interpretation
 - a. (implemented in subsequent slides)

Linkography as a Model of Idea Flow

Adapted from: Linkography: Unfolding the Design Process - Gabriela Goldschmidt

- Linkography represents how conceptual contributions connect, build, and converge over time, rather than treating utterances as independent events
- Originally developed to study design reasoning, capturing moments of expansion, refinement, and synthesis
- Focuses on structural relationships between ideas, not speaking volume or turn-based conversations

Why this matters: Early temporal patterns suggested structured idea flow that requires modeling relationships between ideas, not just behavioral-segmented counts

Mapping Linkography Concepts to Behavioral Annotations

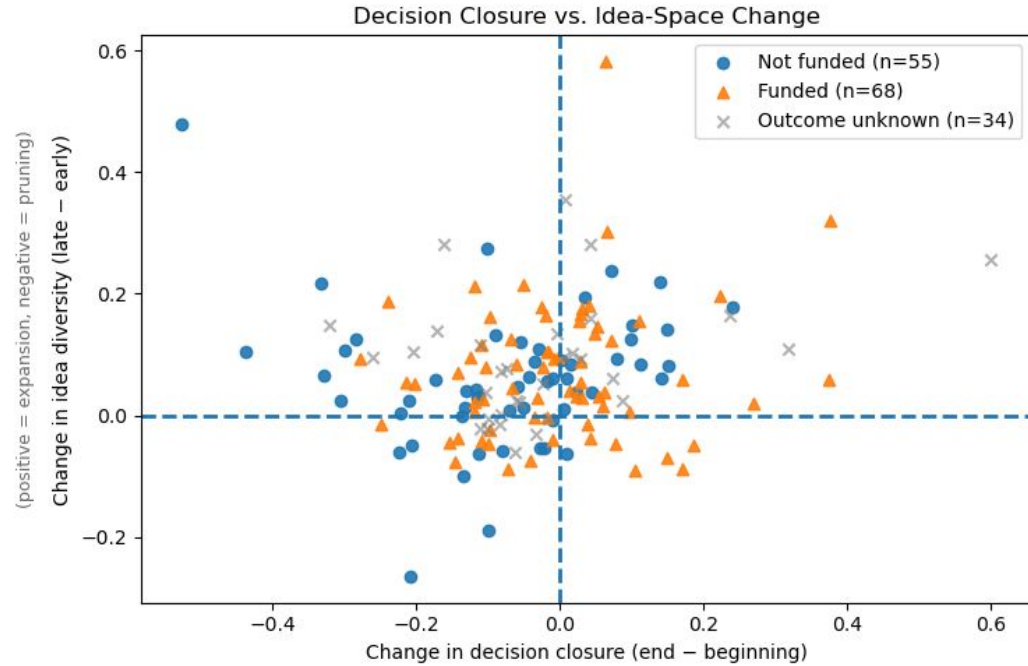
- We map core linkography concepts (e.g., forelinks, backlinks, critical moves) to existing SCIALOG behavioral annotations
- This enables an approximate representation of idea flow

Linkography concept	STM annotation(s)	How is it detected	Why it matters
Forelink	Knowledge Sharing, Idea Management	Consecutive ideation segments	Progressive idea development
Backlink	Evaluation Practices	Explicit reference to prior statement	References to prior ideas
Critical Move	Integration Practices, Coordination & Decision	Moments of idea integration	Decision or transition points
Chunk	Repeated Knowledge Sharing / Idea Management clusters	Short sequences of repeated idea-focused actions	Sustained shared focus
"Idea"	Idea Management, Knowledge Sharing, Integration Practices	The topical or conceptual concept being proposed, referenced, refined, or evaluated within a conceptual contribution	Ideas are the semantic material that conceptual contributions act upon and link over time; tracking them enables proposal convergence, divergence, and refinement across the meeting

Late-Stage Convergence Differentiates Successful Proposals

Slide 10 - First Attempt

- Funded teams show late-stage idea narrowing coupled with decision closure
- Non-funded teams tend to maintain or expand idea space without comparable closure



Across sessions, successful proposals are associated with late-stage pruning of idea-space alongside increased decision and commitment behavior. In contrast, unsuccessful or unresolved sessions tend to retain higher late-stage idea diversity without decisive convergence.

Slide 10 - Explanation (Part 1)

I started by treating the annotation codes as basic behavioral building blocks, rather than the labels they were in the codebook. This is because I kept relaying back to the idea of team formation, rather team funding as the finalized output -- that's what was on my mind at the time cause everything keeps tying back to that. Cause I was trying to describe at the certain parts, what gets a team funded?

So then trying to piece those parts together because there could be up to three codes with one utterance, I was trying to make it a bit simpler on myself. As I also knew the annotation codes wouldn't be able to tell the story by themselves - I also ended up using the respective transcripts to end up checking it as well. For this, I ended up grouping the original annotation codes by the cognitive function they serve in a team's reasoning process, rather than focusing on the duration or frequency a specific annotation code shows up.

That led to this idea of something like the following (just to keep my head intact, not saying all meetings followed this structure to a tee):

****Seeding/Expansion --> Stress-testing --> Synthesis --> Commitment****

I had grouped `Knowledge Sharing` and `Information Seeking` under a `Expansion/Seeding` category because both of these behaviors introduce or clarify information that expands the set of possible ideas under consideration. `Knowledge Sharing` ends up bringing in content into the conversation; `Information Seeking` deals with missing information or clarification.

I had grouped `Evaluation Practices` on its own under a `Stress-testing` category because it does not expand the idea space, but rather compares, critiques or constrains the existing ideas. This all represents `stress-testing`, which alternatives are weighed

Slide 10 - Explanation (Part 2)

I had grouped `Integration Practices` as a `synthesis` category because it involves the combining, reconciling, or reframing multiple ideas into a more robust structure. This actively constructs something new from existing ideas.

I had finally grouped `Coordination and Decision Practices` as a `commitment` behavior. This effectively signals alignment and forward movement; not necessarily an idea exploration. This may include choosing a direction, assigning tasks, defining next steps, formalizing decisions. Not all of a scientific teams' interactions would show this behavior, but this behavior was the last "part" that still needed to be included.

`Idea Management` was given its own category of `IDEA\_MGNT` because it captures the meta-behavior about how ideas are tracked, referenced, or organized over time. This also may include behaviors such as labeling ideas, revisiting earlier suggestions, considering alternative ideas, transitioning between ideas, or even just the continuity throughout the meeting -- keeping the structure together.

For this, `Relational Climate` was never filtered into one of these categories, or given its own, because it was more so about the emotional aspect of these scientific team meetings, cause I again, kept tying all of this to the proposal outcomes.

After I had defined all of these higher-level behavioral, I then incorporated all of this back into the temporal third segments (beginning, middle, end). The goal was to test whether these grouped behaviors shift over time in ways consistent with convergence. Because of the large amount of meetings and knowing they all would be different in terms of structure, I had focused on how these grouped behaviors transition into one another. This then allowed me to find out which behavioral states, trying to keep in-tact an idea-state flow rather than isolating the individual annotation codes. For this I treated `IDEA\_MGMT` as a way to support the transitions without implying convergence (also to keep true to the codebook), to seeing if a relationship with these behaviors hold. This is before tying it to outcomes such as team formation or funding.

Slide 10 - Explanation (Part 3)

Once all these transitions were figured out, I had to figure out whether a `synthesis` or `commitment/closure` reflected genuine idea-level convergence or saying it was the end of a meeting. For these two behaviors (`synthesis` and `commitment/closure`, which involves the behaviors these are associated with in the codebook - see earlier section) I had figured out that these coordination and decision-like behaviors often increase near the end of the meetings due to the facilitation, reporting, or time management - not necessarily due to substantive agreement. I ended up running a false-positive diagnostic analysis while auditing the transcript for reference and using a lexical pattern detection. I then inspected these segments in context to distinguish when the meeting actually finished (because they ran out of time) compared to a genuine convergence, when people were wrapping up on an idea.

This then led me to figuring out that the narrowing does not rely on the behavioral annotations. This is what led me to figure out whether the diversity of ideas under discussion decreases over time. This led me to using an entropy to measure this conceptual spread. This motivation was because that the genuine convergence (e.g. a best idea, correct idea, final decision, successful outcome, etc.) should not only be reflected in how teams behave, but also to the extent the ideas are actually being considered. I then computed the entropy separately for the first-third and last-third of each meeting. For that, I defined a pruning score as the change in entropy over time -- negative values indicate narrowing of an idea space.

Slide 10 - Explanation (Part 4)

After this part was run, it was figured out that some of the meetings had strong `synthesis` and `commitment/closure` behaviors without a reduction in the entropy. Others showed conceptual narrowing without clear `commitment/closure` behavior. For this part I ended up checking against the transcript to figure out the exact cases and words spoken by the researchers.

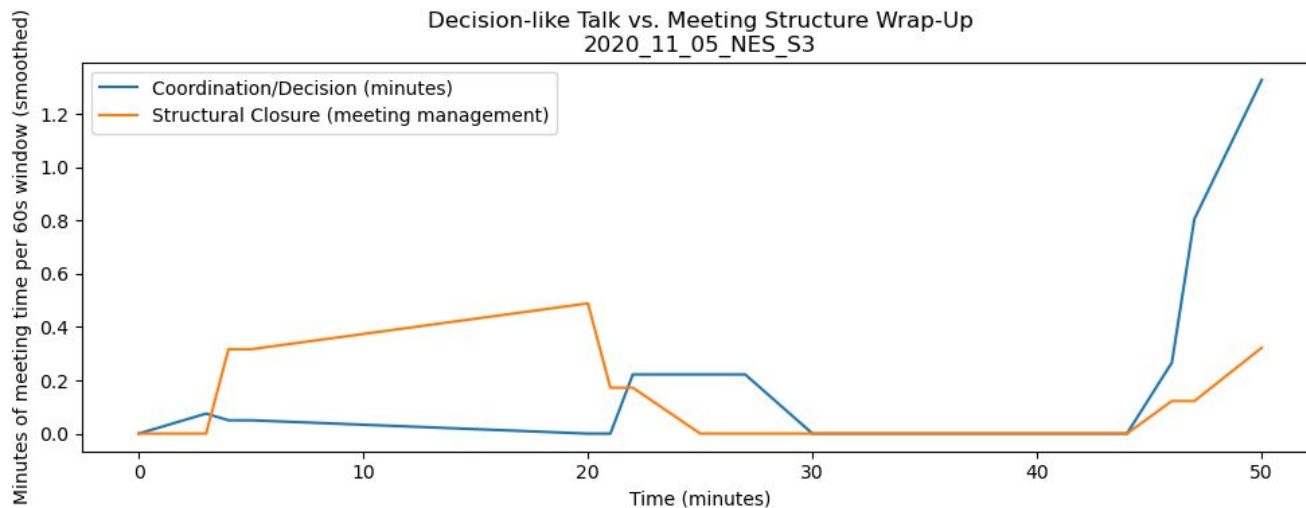
Addressing this limitation, this holds because teams can exhibit `synthesis` and coordination behaviors without adhering to a specific idea, and they can narrow their idea space without signaling commitment through decision-oriented behaviors. This therefore shows that these annotations and transition dynamics can help figuring out where convergence of an idea may be occurring, but insufficient on what the team has converged on. Some sort of semantic tracking would be required to confirm genuine idea-level closure.

So then given all of this, ****to assess decision closure****, I computed late-stage gains in Commitment/Closure relative to early segments, and examined whether these gains occurred after synthesis rather than in isolation. However, because commitment-like behaviors often occur near the end of meetings due to facilitation or reporting, I implemented a false-positive diagnostic using lexical pattern detection. Specifically, utterances containing facilitation or report-out language (e.g., time management, slide preparation, reporting instructions) were flagged via regex-based matching and audited in the transcript. Commitment or synthesis signals overlapping with these patterns were treated as potential structural closure rather than genuine idea-level convergence.

****To assess narrowing independently of behavioral annotations****, I computed idea-space entropy for the first and last thirds of each meeting, treating the distribution of idea states as a proxy for conceptual spread. A pruning score was defined as the change in entropy over time, with negative values indicating narrowing of the idea space. This provided a code-agnostic measure of whether teams reduced the range of ideas under active discussion.

Finally, I compared behavioral convergence signals (`synthesis` and `commitment` gains) with entropy-based narrowing. Meetings where these signals aligned were treated as strong candidates for convergence, while mismatches were examined via transcript inspection. These mismatches revealed that behavioral convergence and conceptual narrowing do not always co-occur, motivating the conclusion that behavioral annotations can localize where convergence may be occurring, but are insufficient to determine what idea a team has converged on without additional semantic tracking.

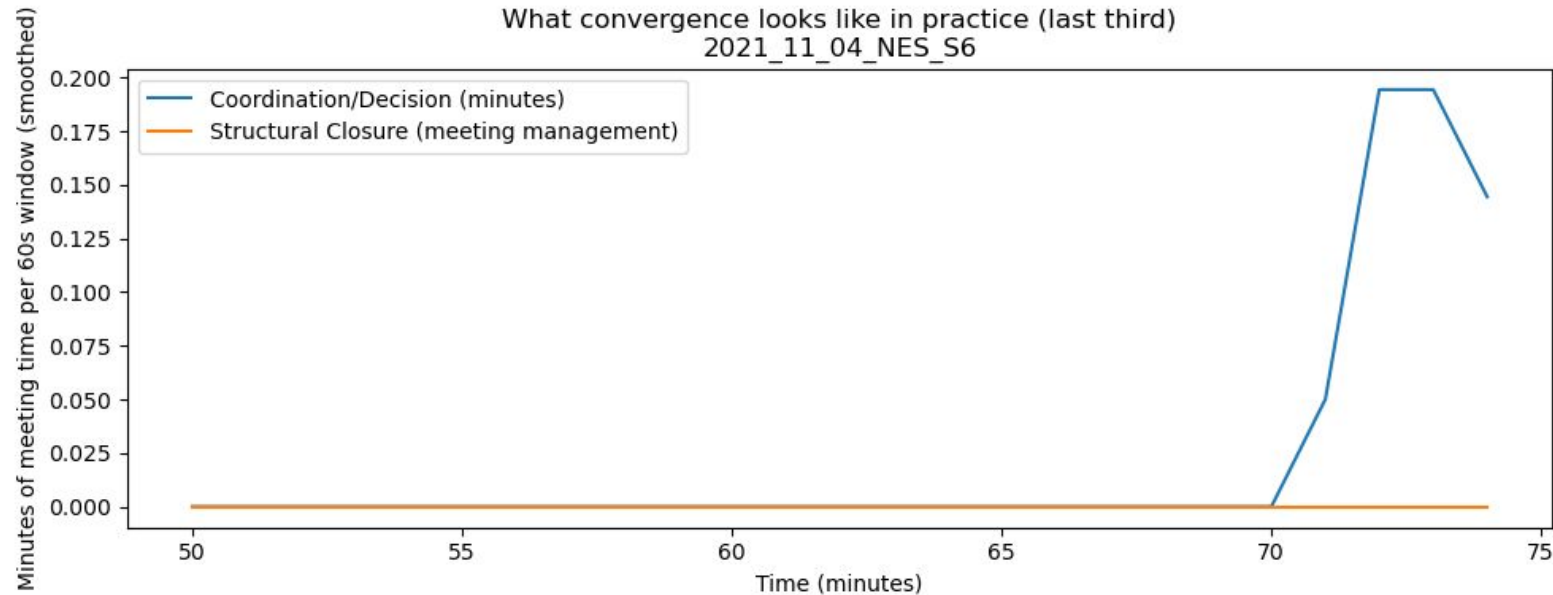
Why Decision $\neq$ Convergence



Quote from the example session (46:26 - 46:48):

“Great. So we’re actually at about eight minutes from close here... I’d like to give Xiao some help with reporting... do you want to share your screen...”

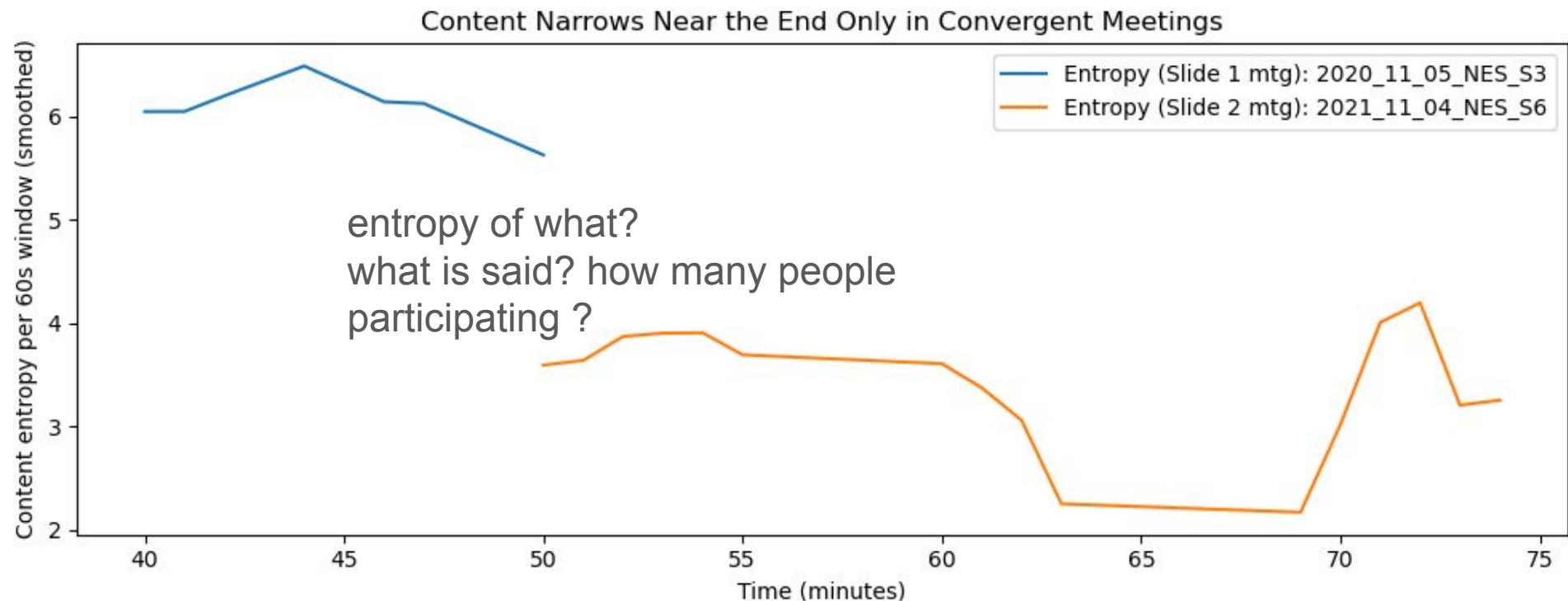
What Convergence Looks Like in Practice



Quote from the example session (72:42 - 73:08):

“Okay, yeah, I agree with you Will... we need to be more specific about what we’re going to do and have a clear plan for how we’re going to achieve the goals”

Only Some End-of-Meeting Decisions Narrow the Discussion



In meetings with convergence, entropy drops near the end, indicating that the discussion narrows onto a small subset of ideas.

Convergence Defined by Co-occurring Signals

- **Unit of Analysis:** 60-second time bins of utterances within a meeting.
- **What's Tracked:** Annotation Codes (Coordination/Decision Practices) treated as categorical states.
- **Entropy Used:** Shannon entropy over the distribution of active annotation codes within each 60-second bin.

When Do We Call a Moment ‘Convergent’?

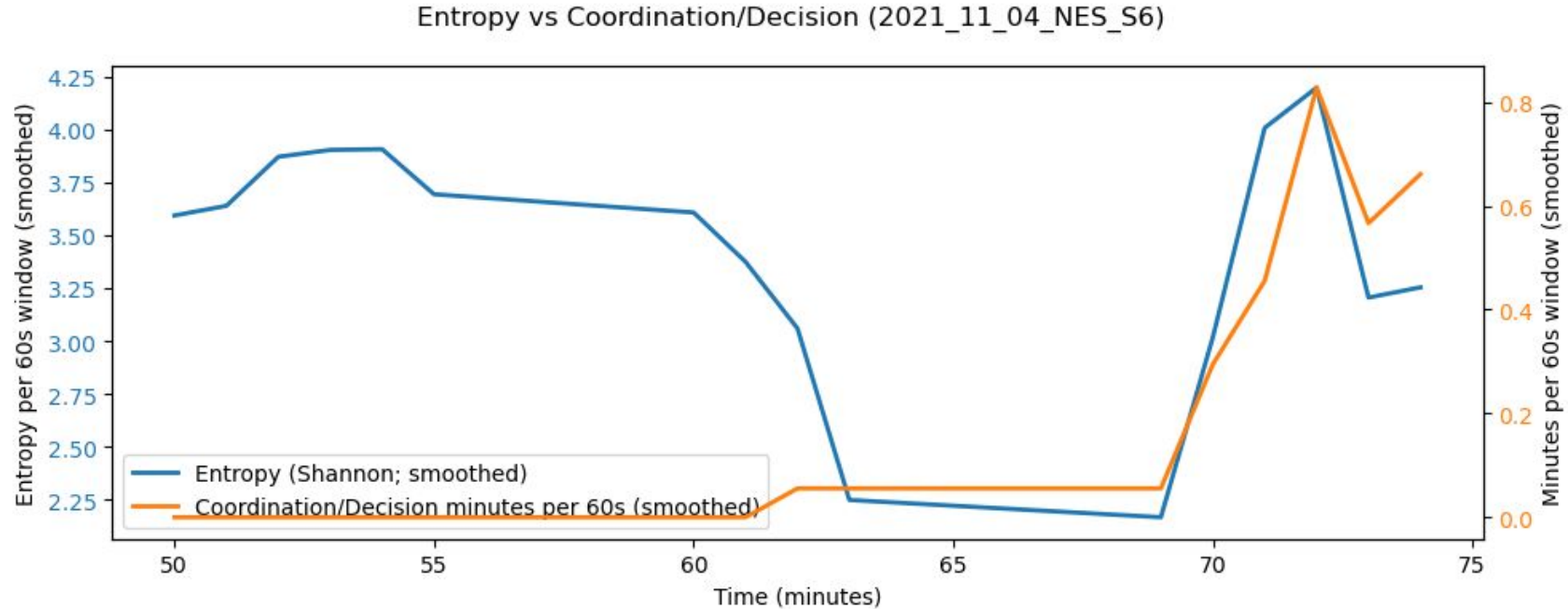
From the two session examples:

- Coordination & Decision Practices are sustained
- Content entropy remains low, indicating focus on fewer ideas

Why Use Entropy as a Signal of Convergence?

- Putman & Paulus (2009): Groups generate many ideas, then converge on a small subset for implementation
- Kirschner et al. (2006): Collaboration involves moving from diverse perspectives toward shared common ground
- Mondada (2006): Interactional practices activity closure before explicit decisions are stated
- Wiltshire et al. (2017): Declines in communication entropy mark transitions toward more focused problem-solving.

Convergence Appears Before Decisions Are Stated



NES_S6: convergent case
(second example)